WORKED SOLUTION

STEP II 2005, Question 13

Mechanics • Poisson distribution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q13

The key idea

Each sampled page either has fewer than two errors or it doesn't, and the pages behave independently, so Y counts successes in n independent trials: it is binomial. Everything after that is about what happens when λ is small, and the tool for that is the Maclaurin series of $\mathrm{e}^{-\lambda}$ (or of $\ln(1+\lambda)$), keeping only the first term that doesn't cancel.

Solution

The distribution of Y

Let X be the number of errors on a particular page, so $X \sim \mathrm{Po}(\lambda)$. Call a page *good* if $X \leq 1$. Then

$$\mathrm{P}(\text{good}) = \mathrm{P}(X=0) + \mathrm{P}(X=1) = \mathrm{e}^{-\lambda} + \lambda\mathrm{e}^{-\lambda} = (1+\lambda)\mathrm{e}^{-\lambda} = p.$$

The sample consists of n different pages. The error counts on different pages are independent and all have the same $\mathrm{Po}(\lambda)$ distribution, so whichever n pages are chosen, the events "page j is good" are independent, each with probability p. (The condition $n \ll N$ just reassures us that there are plenty of pages to sample from; it isn't needed for the independence, which comes from the model.)

So Y is the number of successes in n independent trials with success probability p. A particular set of k good pages and $n-k$ bad ones has probability $p^k q^{n-k}$, and there are $\binom{n}{k}$ ways to choose which k pages are good, so

$$\mathrm{P}(Y=k) = \binom{n}{k} p^k q^{n-k}, \qquad k = 0, 1, \ldots, n,$$

with $p = (1+\lambda)\mathrm{e}^{-\lambda}$ and $q = 1-p$, as required. That is, $Y \sim \mathrm{B}(n, p)$.

Part (i)

Expand $e^{-\lambda} = 1 - \lambda + \frac{1}{2}\lambda^2 - \frac{1}{6}\lambda^3 + \cdots$ and multiply by $1 + \lambda$:

$$\begin{aligned} p = (1+\lambda)e^{-\lambda} &= (1 - \lambda + \tfrac{1}{2}\lambda^2 - \tfrac{1}{6}\lambda^3 + \cdots) + (\lambda - \lambda^2 + \tfrac{1}{2}\lambda^3 - \cdots) \\ &= 1 - \tfrac{1}{2}\lambda^2 + \tfrac{1}{3}\lambda^3 - \cdots . \end{aligned}$$

The terms in λ cancel, so

$$q = 1 - p = \tfrac{1}{2}\lambda^2 - \tfrac{1}{3}\lambda^3 + \cdots \approx \tfrac{1}{2}\lambda^2$$

when λ is small, since the neglected terms are smaller than $\frac{1}{2}\lambda^2$ by a factor of order λ.

Ask yourself Why isn't $q \approx \lambda$, as the first term of $e^{-\lambda}$ might suggest?

Because the $-\lambda$ from $e^{-\lambda}$ is exactly cancelled by the $+\lambda$ from multiplying by $1 + \lambda$. A page is bad only if it has at least two errors, and $P(X \geq 2) \approx P(X = 2) = \frac{1}{2}\lambda^2 e^{-\lambda} \approx \frac{1}{2}\lambda^2$. That is a nice independent confirmation of (i).

Part (ii)

$P(Y = n) = p^n$, and we need the largest n with $p^n \geq 1 - \lambda$. Both sides are positive (for $0 < \lambda < 1$), and ln is increasing, so this is equivalent to

$$n \ln p \geq \ln(1 - \lambda).$$

Now $\ln p = \ln(1 + \lambda) - \lambda = (\lambda - \frac{1}{2}\lambda^2 + \cdots) - \lambda \approx -\frac{1}{2}\lambda^2$, and $\ln(1 - \lambda) \approx -\lambda$. Since $\ln p < 0$, dividing by it reverses the inequality:

$$n \leq \frac{\ln(1-\lambda)}{\ln p} \approx \frac{-\lambda}{-\frac{1}{2}\lambda^2} = \frac{2}{\lambda}.$$

So the largest such n is approximately $2/\lambda$.

Equivalently, using (i): $p^n = (1-q)^n \approx 1 - nq \approx 1 - \frac{1}{2}n\lambda^2$, and $1 - \frac{1}{2}n\lambda^2 \geq 1 - \lambda$ exactly when $n \leq 2/\lambda$. The step $(1-q)^n \approx 1 - nq$ needs nq small; here $nq \approx \frac{2}{\lambda} \cdot \frac{1}{2}\lambda^2 = \lambda$, which is small, so the approximation is self-consistent.

Part (iii)

Since $Y > 1$ implies $Y > 0$,

$$P(Y > 1 \mid Y > 0) = \frac{P(Y > 1)}{P(Y > 0)} = \frac{P(Y > 0) - P(Y = 1)}{P(Y > 0)} = 1 - \frac{npq^{n-1}}{1 - q^n}.$$

(Here $n \geq 2$ matters: it makes $Y > 1$ a possible event and the power q^{n-1} at least q.) Now let λ be small with n fixed. Then $q \approx \frac{1}{2}\lambda^2$ is small, so $p \approx 1$ and $1 - q^n \approx 1$, giving

$$P(Y > 1 \mid Y > 0) \approx 1 - nq^{n-1} \approx 1 - n\left(\frac{\lambda^2}{2}\right)^{n-1},$$

using (i) in the last step. This is the required result.

Checks

- A simulation of 200,000 samples of $n = 5$ Poisson pages with $\lambda = 0.8$ gave relative frequencies for $Y = 0, \ldots, 5$ agreeing with $\mathrm{B}(5, p)$ to within 0.002.
- For $\lambda = 0.01$, $q = 4.967 \times 10^{-5}$, against $\frac{1}{2}\lambda^2 = 5 \times 10^{-5}$.
- For $\lambda = 0.01$, the exact largest n with $p^n \geq 0.99$ is 202, against $2/\lambda = 200$. (Keeping one more term in each series gives $2/\lambda + \frac{7}{3}$, so the error stays bounded while $2/\lambda$ grows.)
- For $\lambda = 0.01$, $n = 3$: $1 - \mathrm{P}(Y > 1 \mid Y > 0) = 7.40 \times 10^{-9}$ exactly, against $3(\lambda^2/2)^2 = 7.5 \times 10^{-9}$.

Common slips

- Taking "fewer than two errors" as $X = 0$ only, or as $X \leq 2$.
- Stopping the expansion of $(1 + \lambda)\mathrm{e}^{-\lambda}$ at the linear term and concluding $q \approx 0$: the first surviving term is the quadratic one.
- Forgetting to reverse the inequality when dividing by $\ln p < 0$ in (ii).

Beyond STEP

Part (ii) is an example of asymptotic analysis: the condition is controlled by the product $n\lambda^2$, so the natural scale for n is $1/\lambda$, and quantities like p^n behave like $\mathrm{e}^{-n\lambda^2/2}$. The same reasoning underlies the Poisson approximation to the binomial and many "rare events" limits in probability. Grimmett and Welsh, *Probability: An Introduction*, develops these limit theorems carefully.

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